

Carbon and its Compounds (Set A) — MCQ Worksheet

Science • Chapter 4 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

- Q1.** Carbon forms a large number of compounds mainly because of:
- (A) Tetravalency and catenation (B) High atomic number
(C) Metallic nature (D) Inert nature
- Q2.** The valency of carbon is:
- (A) 1 (B) 2
(C) 3 (D) 4
- Q3.** The bonds in carbon compounds are mostly:
- (A) Ionic (B) Covalent
(C) Metallic (D) Hydrogen
- Q4.** Saturated hydrocarbons are also called:
- (A) Alkenes (B) Alkynes
(C) Alkanes (D) Aromatic compounds
- Q5.** The general formula of alkanes is:
- (A) C_nH_{2n} (B) C_nH_{2n+2}
(C) C_nH_{2n-2} (D) C_nH_n
- Q6.** Ethene has the formula:
- (A) C_2H_2 (B) C_2H_4
(C) C_2H_6 (D) CH_4
- Q7.** The functional group $-COOH$ is called:
- (A) Aldehyde (B) Ketone
(C) Carboxylic acid (D) Alcohol
- Q8.** The IUPAC name of CH_3CH_2OH is:
- (A) Methanol (B) Ethanol
(C) Propanol (D) Methanal
- Q9.** A homologous series differs by:
- (A) $-CH_3$ (B) $-CH_2-$
(C) $-OH$ (D) $-COOH$
- Q10.** Burning of saturated hydrocarbons produces:
- (A) Sooty yellow flame (B) Clean blue flame
(C) No flame at all (D) Green flame
- Q11.** Unsaturated hydrocarbons undergo:
- (A) Substitution (B) Addition
(C) Elimination only (D) No reactions
- Q12.** The catalyst used in hydrogenation of vegetable oils is:
- (A) Pt (B) Ni
(C) Fe (D) Cu
- Q13.** Methane reacts with chlorine in sunlight to give methyl chloride. This is a:
- (A) Combination reaction (B) Substitution reaction
(C) Addition reaction (D) Decomposition reaction
- Q14.** Glacial acetic acid is:
- (A) Dilute ethanoic acid (B) Pure CH_3COOH
(C) Aqueous solution of HCl (D) Mixture of acid and water
- Q15.** Vinegar is approximately:

- (A) 100% acetic acid
(C) Pure ethanol

- (B) 5–8% acetic acid
(D) Pure water

Q16. Esterification reaction is:

- (A) Alcohol + Acid $\rightarrow$ Ester + Water
(C) Ester + Water $\rightarrow$ Alcohol + Acid

- (B) Alcohol + Water $\rightarrow$ Ester
(D) Acid + Acid $\rightarrow$ Ester

Q17. The product formed when ethanol reacts with sodium is:

- (A) Sodium ethoxide + H₂
(C) Ethyl acetate

- (B) Sodium acetate
(D) Methane

Q18. Ethanoic acid + sodium bicarbonate gives:

- (A) Salt + water
(C) Only CO₂

- (B) Salt + water + CO₂
(D) Ester

Q19. The hydrophobic part of a soap molecule is:

- (A) The –COO[–] end
(C) The Na⁺ ion

- (B) The long hydrocarbon chain
(D) The water layer

Q20. Soaps form micelles in water. Inside a micelle:

- (A) Heads point inward
(C) All ions are arranged randomly

- (B) Tails point inward and trap oily dirt
(D) There is no oil at all

Q21. Hard water contains ions of:

- (A) Na⁺ and K⁺
(C) Fe³⁺ only

- (B) Ca²⁺ and Mg²⁺
(D) H⁺ only

Q22. Synthetic detergents work in hard water because:

- (A) They form scum easily
(C) They are stronger acids

- (B) Their Ca/Mg salts are soluble
(D) They contain soap

Q23. The IUPAC name of CH₃COOH is:

- (A) Methanoic acid
(C) Propanoic acid

- (B) Ethanoic acid
(D) Butanoic acid

Q24. Diamond and graphite are:

- (A) Different compounds of carbon
(C) Isotopes of carbon

- (B) Allotropes of carbon
(D) Mixtures of carbon

Q25. The number of covalent bonds in a methane molecule is:

- (A) 2
(C) 4

- (B) 3
(D) 5